

9. Aluminium metal is produced industrially using electrolysis, due to the relatively high stability of aluminium oxide.

The overall equation for the reaction is



- (a) Describe, in terms of electrons, the reduction and oxidation occurring during the electrolysis.

[2]

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- (b) (i) Aluminium ore contains 36 % aluminium oxide, Al_2O_3 .

Calculate the mass of aluminium oxide in 500 tonnes of the ore.

[1]

Mass = tonnes

- (ii) Calculate the mass of aluminium metal that could be produced from this mass of aluminium oxide.

[3]

$$A_r(\text{Al}) = 27$$

$$A_r(\text{O}) = 16$$

Mass = tonnes

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END OF PAPER

